COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/09/2026	**DATE SUBMITTED:**	10/09/2026
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ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

inventory.ini

```
  GNU nano 8.7.1
[web_servers]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1
centos-node ansible_host=192.168.56.107 ansible_user=sysadleroy1

[workstations]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1

[db_servers]
centos-node ansible_host=192.168.56.107 ansible_user=sysadleroy1

[file_servers]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1
```

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it “files.” Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.



2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

```
  - name: copy default html file for site
    tags: apache, apache2, httpd
    copy:
      src: files/default_site.html
      dest: /var/www/html/index.html
      owner: root
      group: root
      mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

```
leroy1@leroyque: ~/CPE212_QUE
~/CPE212_QUE

           ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centos-node]

TASK [Mariadb - Restarting/Enabling] ****************************************
changed: [centos-node]

PLAY [file_servers] *********************************************************

TASK [Gathering Facts] ******************************************************
ok: [ubuntu-node]

TASK [install samba package] ************************************************
changed: [ubuntu-node]

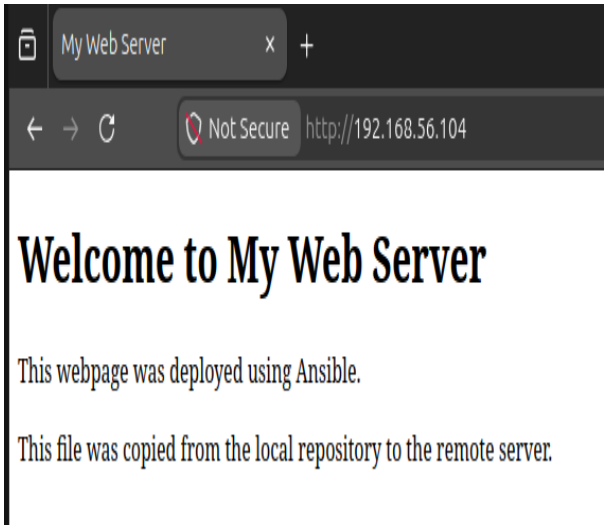
PLAY RECAP ******************************************************************
centos-node                : ok=9    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
ubuntu-node                : ok=7    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0

leroy1@leroyque:~/CPE212_QUE$
```

Analysis: The ansible has connected the VMs we are targeting. It transfers the file from the control node which is the default_site.html to /var/www/html/index.html on both the managed nodes we have. It also sets the ownership to root:root with permissions 0644.

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

```
leroy1@leroyque:~/CPE212_QUE$ ssh leroy1@192.168.56.104 "cat /var/www/html/index.html"
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
leroy1@leroyque:~/CPE212_QUE$ ssh sysadleroy1@192.168.56.107 "cat /var/www/html/index.html"
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>
    <h1>Welcome to My Web Server</h1>
    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```





Analysis: We have successfully displayed the site on both the managed nodes.

5. Sync your local repository with GitHub and describe the changes.



Analysis: The directory 'files' has been added which includes the default_site.yml alongside the modification for inventory.ini and site.yml.

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play:

```
- hosts: workstations
  become: true
  tasks:
    - name: install unzip
      ansible.builtin.package:
        name: unzip

    - name: install terraform
      ansible.builtin.unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
        dest: /usr/local/bin
        remote_src: yes
        mode: '0755'
        owner: root
        group: root
```

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.

```
  GNU nano 8.7.1
[web_servers]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1
centos-node ansible_host=192.168.56.107 ansible_user=sysadleroy1

[workstations]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1

[db_servers]
centos-node ansible_host=192.168.56.107 ansible_user=sysadleroy1

[file_servers]
ubuntu-node ansible_host=192.168.56.104 ansible_user=leroy1
```

3. Run the playbook. Describe the output.

```
TASK [Gathering Facts] *********************************************************************************************************************************************************************
ok: [centos-node]

TASK [install mariadb package (CentOS)] ****************************************************************************************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/leroy1/CPE212_QUE/site.yml:88:11

86       name: mariadb-server
87       state: latest
88     when: ansible_distribution == "CentOS"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

ok: [centos-node]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/leroy1/CPE212_QUE/site.yml:95:11

93       name: mariadb-server
94       state: latest
95     when: ansible_distribution == "Ubuntu"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centos-node]

TASK [Mariadb - Restarting/Enabling] *******************************************************************************************************************************************************
changed: [centos-node]

PLAY [file_servers] ************************************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************************************
ok: [ubuntu-node]

TASK [install samba package] ***************************************************************************************************************************************************************
ok: [ubuntu-node]

PLAY RECAP *********************************************************************************************************************************************************************************
centos-node                : ok=9    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
ubuntu-node                : ok=10   changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Analysis: The unzip package was installed first to make sure that we have the archive extraction tools. After that, the ansible will fetch the terraform.zip file then unzip it to /usr/local/bin, and then 0755 executable permission were assigned with root ownership.

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.

```
leroy1@leroyque:~$ terraform -v
Terraform v0.12.28

Your version of Terraform is out of date! The latest version
is 1.16.2. You can update by downloading from https://www.terraform.io/downloads.html
```

Analysis: I ssh into the ubuntu node and then check whether terraform was installed by checking its version.

Task 3: Create roles

1. Edit the site.yml. Configure roles as fo lows: (make sure to create a copy of the old site.yml file because you wil be copying the specific plays for a l groups)

```
  GNU nano 8.7.1
- hosts: all
  become: true
  pre_tasks:
    - name: update repository index (CentOS)
      tags: always
      ansible.builtin.dnf:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      ansible.builtin.apt:
        update_cache: yes
      changed_when: false
      when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

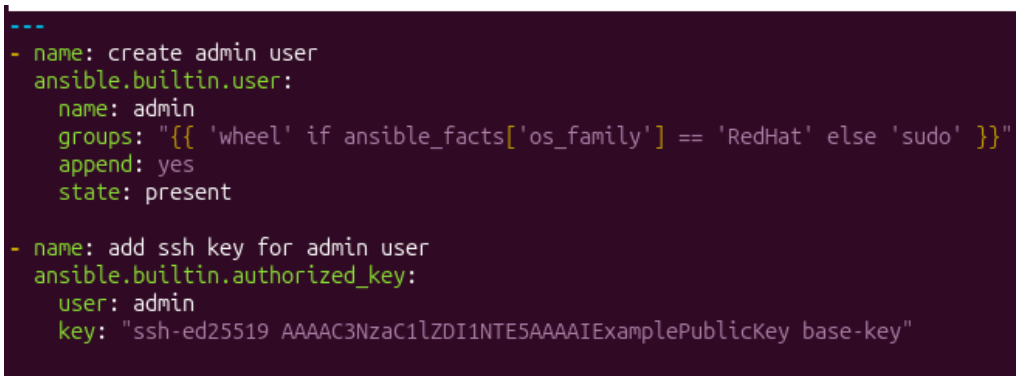
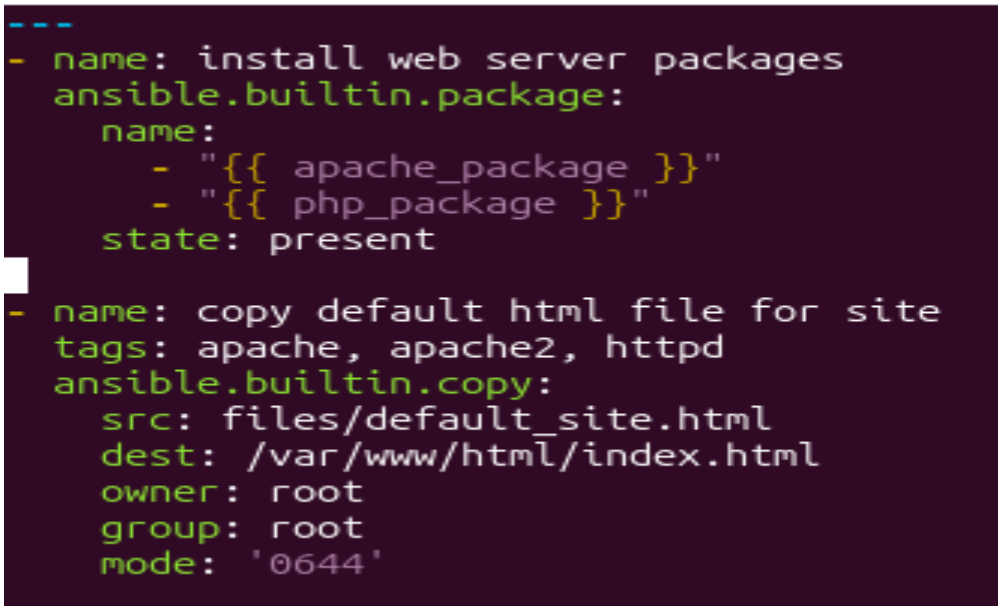
- hosts: db_servers
  become: true
  roles:
    - db_servers

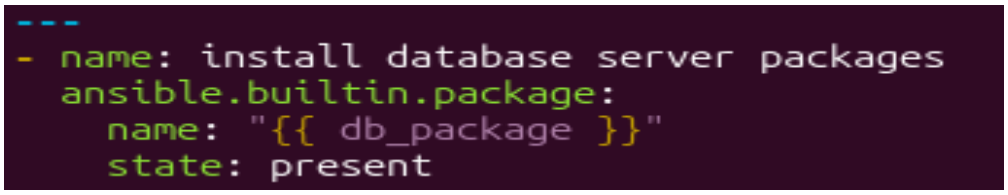
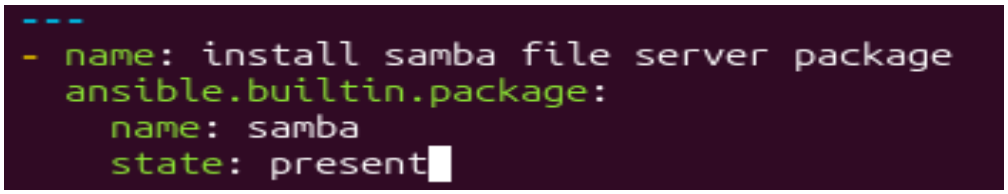
- hosts: file_servers
  become: true
  roles:
    - file_servers
```

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

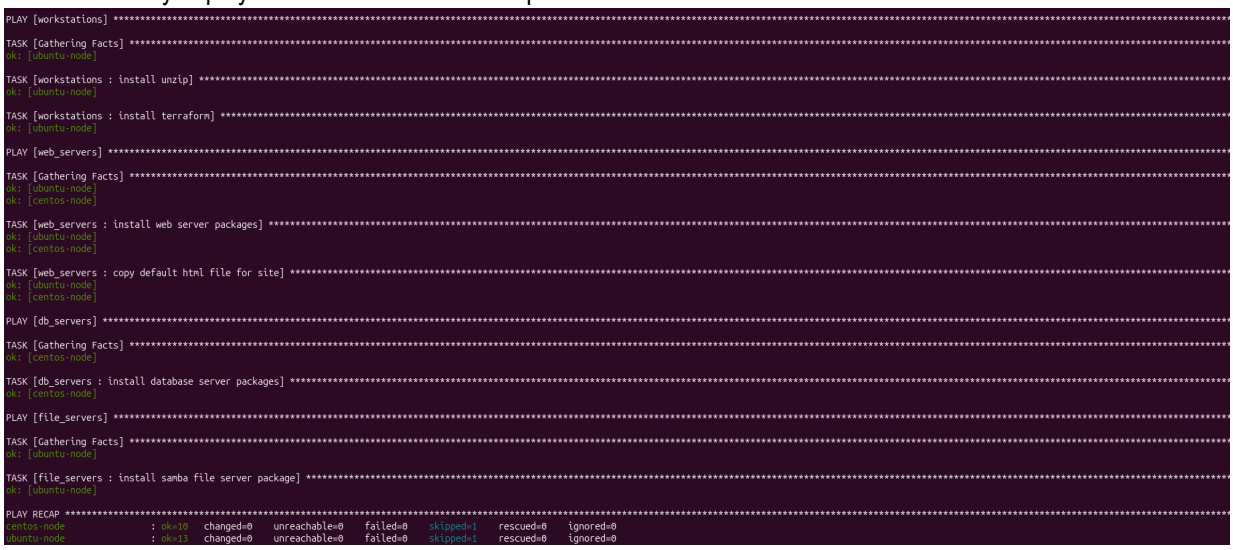
```
leroy1@leroyque:~/CPE212_QUE$ mkdir -p roles/{base,web servers,file servers,db servers,workstations}/tasks
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

Directory	Content
roles/base/tasks/main.yml	
roles/workstations/tasks/main.yml	
roles/web_servers/tasks/main.yml	

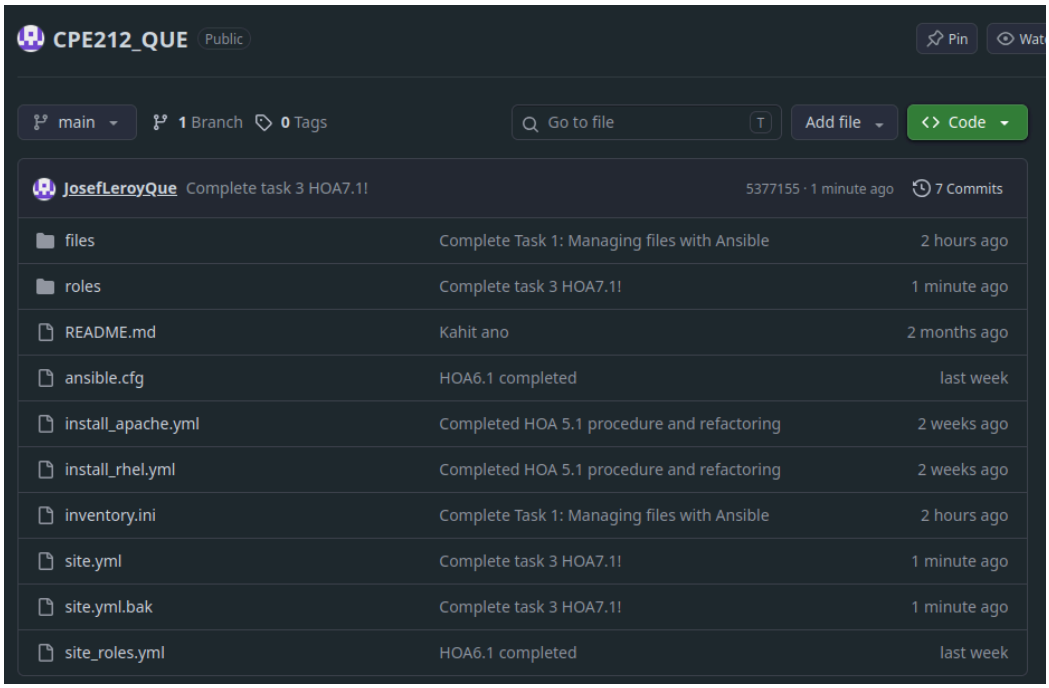
roles/db_servers/tasks/main.yml	
roles/file_servers/tasks/main.yml	

4. Run the site.yml playbook and describe the output.



Analysis: All of the requested packages, role assignments, and configurations were satisfied and completed on the target nodes. This is also a proof of synchronization across the 3 VMs.

Git commit:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?
   - Creating roles allows you to manage your playbooks in a modular fashion. Thereby enabling the sharing of the playbooks over different Operating Systems.
2. What is the importance of managing files?
   - By managing files, it allows one to watch over the distribution of configuration files. Thus eliminating the need for manual copying.

Conclusions/Learnings:

In this laboratory activity, I have understood and learned how to set up automatic configuration of servers on different versions of Linux through the use of Ansible. Moreover, by separating a large playbook into manageable roles, I was able to significantly organize the code in an orderly fashion. Besides that, I had the experience of dealing with the OS-specific discrepancies of naming packages, managing group permissions, etc. Most importantly, after achieving zero failures and zero changes, the significance of the role of the idempotency of administration was highlighted.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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